Finding the best feasible bundle

Lecture notes

What this lecture is about. Last lecture ended with a picture: the best feasible bundle is where the highest reachable indifference curve touches the budget line. Today we go slowly. First the utility functions, one form at a time. Then we look at four kinds of preferences on one budget and see that the best bundle can sit at a corner, at a kink, or where a curve just touches the line. Then we walk along the budget line to see *why* the touching point is best. Then we draw the same idea with arrows, and only then write it as algebra (the Lagrangian). We end with corners and kinks.

One budget for §§2–6. Prices $p_x = 3$, $p_y = 1$, money $B = 60$:

$$3x + y = 60.$$

1 Utility functions, one form at a time

Utility function

A utility function gives every bundle a number, so that a better bundle gets a higher number and equally good bundles get the same number. An indifference curve is all bundles with the same number. Only the order of the numbers matters (ordinal): $2u + 5$ or u^3 describe the same preferences.

form	formula	indifference curves
linear (perfect substitutes)	$u = a\,x + b\,y$	straight lines; the MRS is a/b everywhere
Leontief, $a = b = 1$ (perfect complements)	$u = \min\{x, y\}$	L-shapes, kinks on the line $x = y$
Leontief (the general recipe)	$u = \min\{a\,x, b\,y\}$	L-shapes, kinks on the line $a\,x = b\,y$
Cobb-Douglas	$u = x\,y$ (or $x^\alpha y^\beta$)	smooth, bowed in
quasi-linear	$u = \alpha x + y^\beta$, e.g. $2x + \sqrt{y}$	each curve is the same curve shifted; bowed in for $\beta < 1$
quasi-linear, y squared	$u = x + y^2/60$	shifted copies, but bowed *out*

Perfect complements are the 1 : 1 Leontief ($a = b = 1$). Example of a recipe: every coffee needs exactly 2 sugars, $u = \min\{2x, y\}$, with kinks on $y = 2x$.

A number for every bundle. With $u = x\,y$: $(2, 8)$, $(4, 4)$ and $(8, 2)$ all get 16, so they lie on one indifference curve; $(4, 8)$ and $(8, 4)$ get 32, on a higher one. Double every number, $v = 2xy$: 16

becomes 32 and 32 becomes 64, but the curves do not move and the order stays. Only the order matters: utility is *ordinal*.

Same bundle, different preferences. On the parameter sliders every curve goes through the same bundle $A = (10, 10)$: each curve is a *different person*. One person's indifference curves never cross; different people's can. Ask each person: one more x, how much y would you give? That is the MRS at A:

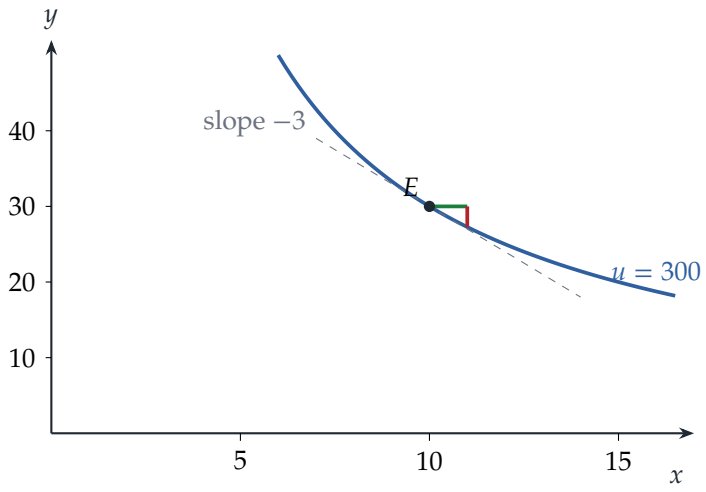
$$\text{substitutes: } \frac{a}{b}, \qquad \text{Cobb-Douglas: } \frac{\alpha}{\beta} \cdot \frac{y}{x}, \qquad \text{quasi-linear: } \frac{\alpha}{\beta} y^{1-\beta}.$$

For perfect complements: on the flat arm nothing (extra x alone adds nothing), on the steep arm the spare y down to the kink. A steeper curve through A means a person who likes x more.

Marginal utility and the MRS from one curve

Marginal utility. With $u = x\,y$ at $A = (5, 45)$: $u = 225$. One more x, the same y: $6 \cdot 45 = 270$, a gain of 45. Why 45? One more x adds one more "y": at $C = (15, 15)$, $16 \cdot 15 = 240$, a gain of $15 = y$. So $MU_x = y$, and in the same way $MU_y = x$ (at A: $5 \cdot 46 = 230$, a gain of 5).

The MRS by staying on one curve. Take the curve $u = x\,y = 300$; it passes through $E = (10, 30)$, and on it $y = 300/x$.



Step from $x = 10$ to 11, staying on the curve: $y = 300/11 = 27.27$, so $\Delta y/\Delta x = -2.727$. Smaller steps give -2.970 ($\Delta x = 0.1$) and -2.997 ($\Delta x = 0.01$), going to -3. With calculus:

$$\frac{dy}{dx} = -\frac{300}{x^2} = -\frac{y}{x} = -\frac{30}{10} = -3 \quad \text{at } E.$$

The general way, for any utility function: along the curve utility stays the same, so for small steps the gain from more x equals the loss from less y:

$$MU_x\,dx + MU_y\,dy = 0 \qquad \Longrightarrow \qquad MRS = -\frac{dy}{dx} = \frac{MU_x}{MU_y}.$$

Here $MRS = y/x = 30/10 = 3$: the same answer both ways. At $A = (5, 45)$: $MRS = 45/5 = 9$: I would give 9 y for one more x.

Quasi-linear, the other way: y squared

With $u = x + y^2/60$ (the $1/60$ only makes the picture visible), $MU_x = 1$ and $MU_y = y/30$, so $MRS = 30/y$: the curves are again shifted copies, but now they bow *out*, because more y makes y worth more, not less. On the budget $3x + y = 60$:

bundle	$(20, 0)$	$(50/3, 10)$	$(10, 30)$	$(0, 60)$
u	20	18.33	25	**60**

The curve just touches the line at $y = 10$ (where $30/y = 3$), and that point is the *worst* on the line. The best is the corner $(0, 60)$. Touching is not enough when the curves bow out.

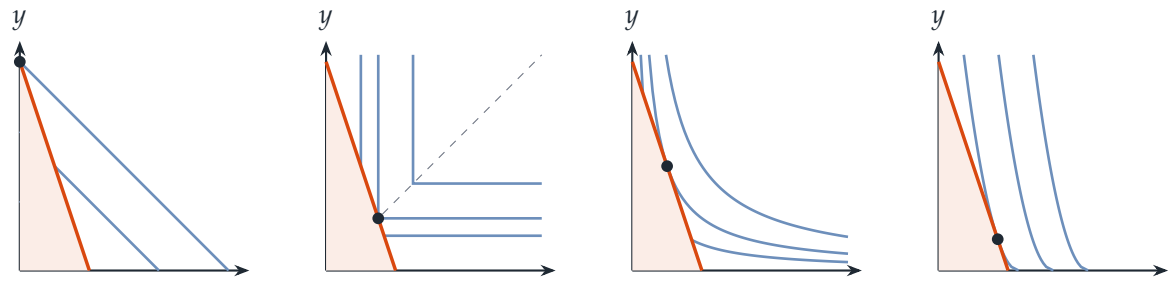
The general idea: y enters through a function. Quasi-linear means $u = \alpha x + v(y)$: perfect substitutes, but y enters through a function $v(y) = y^\beta$. With $\beta = 1$ it is plain perfect substitutes; with $\beta < 1$ each extra y adds less (bowed in, as with the coffees); with $\beta > 1$ each extra y adds *more*. Think of craving (the more you've had, the more you want the next one), binge-watching (the more episodes you have seen, the more you want the next one). The curves bow out, so the best bundle is a corner: all or nothing. (Real addiction runs over time, yesterday to today; Frank ch. 7 shows the other side, tolerance and self-control, pp. 231–242.)

Interior solution, corner solution

An *interior solution*: the best feasible bundle has some of both goods. With smooth curves it is where a curve just touches the budget line. A *corner solution*: all the money goes on one good. With perfect complements the best bundle is at a *kink*: interior (some of both), but not a touching point, because the curve has no single slope there.

2 Four kinds of preferences on one budget

Each panel shows indifference curves (blue) and the same budget line (orange). The black dot is the best feasible bundle.



perfect substitutes $u = x + y$: a **corner**

perfect complements $u = \min\{x, y\}$: a **kink**

Cobb-Douglas $u = xy$: **just touches**

quasi-linear $u = x + 2\sqrt{y}$: **just touches**

1. **Perfect substitutes,** $u = x + y$ (the same energy drink; x: at the U-Bahn kiosk, y: at Billa;

$p_x = 3, p_y = 1$). A can is a can, so one for one: straight lines, MRS = 1. But the kiosk charges three times as much, so you buy them all at Billa: the **corner** $(0, 60)$, $u = 60$.

2. **Perfect complements (Leontief),** $u = \min\{x, y\}$ (hot dogs; x: sausages, y: buns, one each; $p_x = 3, p_y = 1$). A sausage without a bun adds nothing, so only pairs count, and the curves are L-shaped. Their **kinks** sit on the line $x = y$, where nothing is wasted. On the budget: $3x + x = 60$, so 15 hot dogs, $(15, 15)$.

3. **Cobb-Douglas,** $u = xy$ (a party; x: pizza slices, y: soda cans). I like both; the more x I already have compared with y, the less y I give up for one more x, so the curves bow in. A curve **just touches** the line at $(10, 30)$, $u = 300$. This is our point E for the rest of the notes.

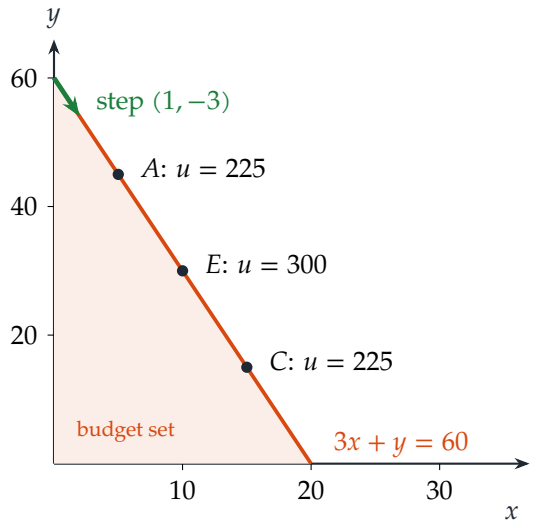
4. **Quasi-linear,** $u = x + 2\sqrt{y}$ (a student's month; x: Mensa lunches, y: coffees; $p_x = 3, p_y = 1$). Every lunch is worth the same; the first coffee wakes you up, the ninth does little. The MRS is $\sqrt{y}$: it depends on y only, so every curve is the same curve shifted sideways. It just touches where $\sqrt{y} = 3$: 9 coffees and 17 lunches, $(17, 9)$, $u = 23$.

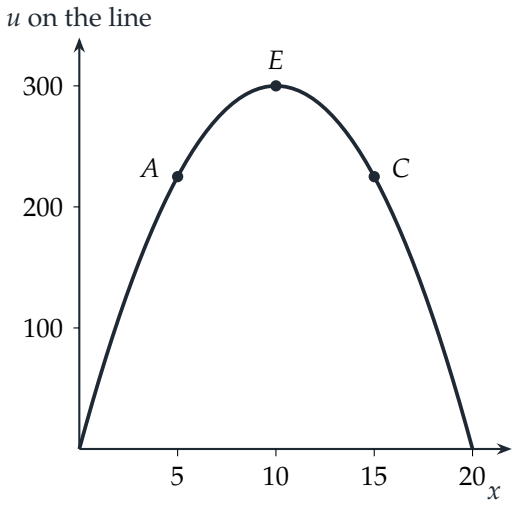
Insight: three shapes of the answer

The best feasible bundle is at a *corner*, at a *kink*, or where a curve *just touches* the line. The rest of today explains why the touching point is best (§§3–6), and why corners and kinks need a different check (§7).

3 Walk along the budget line

From now on until §7: $u = xy$ on $3x + y = 60$.





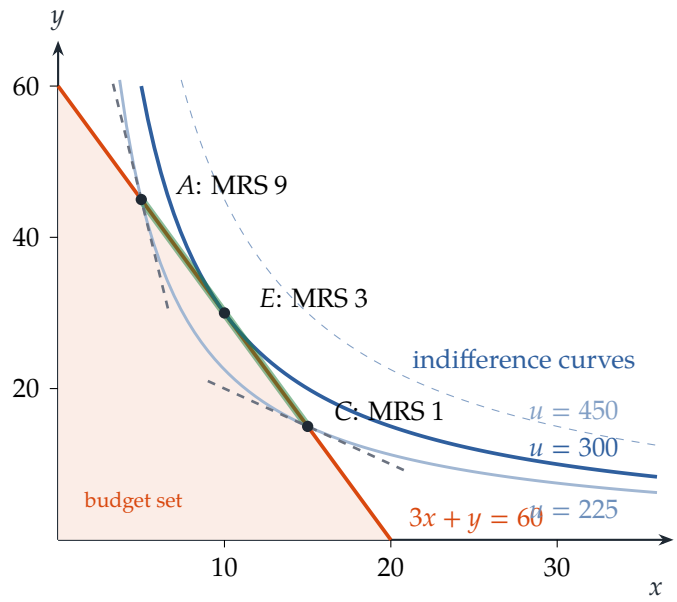
Start at $(0, 60)$: all money on y, but $u = 0 \cdot 60 = 0$. One step right is one more x and three less y; it costs $3 - 3 = 0$, so we stay on the line. At $A = (5, 45)$, $u = 225$. At $E = (10, 30)$, $u = 300$. At $C = (15, 15)$, $u = 225$ again. (C is the bundle the hot-dog person picked in §2; these preferences

are different.) Along the line, utility is

$$u = x\,(60 - 3x),$$

which rises and then falls, like walking over a hill: $0, 225, 300, 225, 0$. **The top of the hill, E, is the best feasible bundle.** In 3D: utility is a hill over the map, and the budget line is a trail across it; walking the trail you go up and then down, and its highest point is E. Two things are fixed, one at a time: along the orange trail *money* is fixed at €60; along the blue height lines *utility* is fixed. At the top the trail is level and touches a height line: seen from above, the curve $u = 300$ just touches the budget line at E, without crossing.

4 At the top, the curve just touches: the same slope



A and C have the same $u = 225$, so they lie on one indifference curve, and it *cuts* the line twice. Every bundle on the line between the two cuts (green) lies on a higher curve; that is the hill of §3, where utility on the line is above 225 between A and C. So neither A nor C can be best. At E the curve $xy = 300$ no longer cuts the line; it only touches it. Higher curves, like 450, miss the line: nothing on them is affordable.

Marginal utility

MU_x is how much u rises with one more unit of x, holding y fixed (a derivative). For $u = xy$: $MU_x = y$ and $MU_y = x$. Check at A: from $(5, 45)$ to $(6, 45)$, u goes from 225 to 270, up by $45 = y$.

The slope of the line is the price ratio, the slope of the curve is the MRS (grey dashes in the picture):

$$\frac{p_x}{p_y} = \frac{3}{1} = 3$$

$$MRS = \frac{MU_x}{MU_y} = \frac{y}{x}: \qquad MRS(A) = \frac{45}{5} = 9, \qquad MRS(C) = \frac{15}{15} = 1, \qquad MRS(E) = \frac{30}{10} = 3.$$

At A I would give 9 y for one more x, and the shop asks only 3: buy more x. At C I would give only 1, the shop asks 3: buy less x. At E my swap rate equals the shop's: no swap helps. **Touching = the same slope:** $MRS = p_x/p_y$.

5 Arrows

5.1 An arrow written as two numbers

Vector and dot product

A *vector* is an arrow written as two numbers: how far right, how far up. $(3, 1)$ is 3 right and 1 up; $d = (1, -3)$ is 1 right and 3 down. Only its direction and length matter, not where we draw it.

The *dot product* multiplies the matching numbers and adds them:

$$(a_1, a_2) \cdot (b_1, b_2) = a_1 b_1 + a_2 b_2.$$

It is 0 exactly when the two arrows stand at a right angle.

5.2 The price arrow stands at a right angle to the budget line

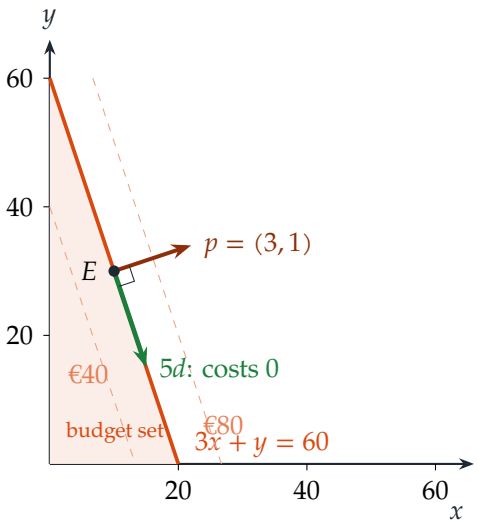
Put the prices into an arrow, $p = (3, 1)$. Dotted with any step (dx, dy), it gives exactly what that step costs:

$$p \cdot (dx, dy) = 3\,dx + 1\,dy.$$

A step along the budget line, $d = (1, -3)$, costs

$$p \cdot d = 3 \cdot 1 + 1 \cdot (-3) = 0,$$

so p stands at a right angle to the line. Think of p as *the cost of a move*: the move $(1, 0)$ costs €3, the move $(0, 1)$ costs €1, and the move $(1, -3)$ costs $3 - 3 =$ €0, a free move. Free moves run along the budget line, and a cost of 0 means a right angle.



Why p is also the fastest way to spend more: the same-spending lines $(3x + y = 40, 60, 80)$ are parallel, and the shortest way from one to the next goes straight across, at a right angle, which is the direction of p. In 3D: spending is a flat roof over the map, the budget line is its €60 height line, and p points straight up the roof.

5.3 The utility arrow: the steepest way up the mountain

Utility is a mountain over the map; the indifference curves are its height lines, like on a hiking map (Frank, App. 3A, Fig. 3A.3). Put the two marginal utilities into an arrow, the *gradient*:

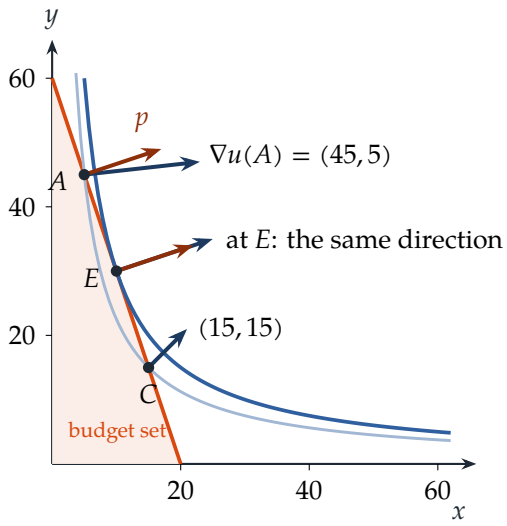
$$\nabla u = (MU_x, MU_y) = (y, x).$$

At A it is $(45, 5)$: one more x adds exactly 45, one more y exactly 5. So a small step changes utility by

$$du = 45\,dx + 5\,dy = \nabla u \cdot (dx, dy),$$

the same shape as its cost $p \cdot (dx, dy)$, with marginal utilities in place of prices. So ∇u is *the gain of a move*, as p is its cost. At E the free move $(1, -3)$ gains $30 - 3 \cdot 10 = 0$: both arrows stand square to the same line, so $\nabla u = \lambda p$, $(30, 10) = 10 \cdot (3, 1)$, that is $MU_x = \lambda p_x$ and $MU_y = \lambda p_y$. Divide by the prices: equal utility per euro, $\lambda = 10$.

Why the gradient stands at a right angle to the indifference curve. Along the curve u does not change, so a small step along the curve has $\nabla u \cdot \text{step} = 0$: a right angle, for the same reason as p and the budget line. And as for p: the shortest way from one height line to the next goes straight across, so the gradient is the steepest way up.



Insight: two arrows, two right angles

p stands at a right angle to the same-spending line; ∇u stands at a right angle to the same-utility curve. At A they point different ways, so the curve and the line cross: A is not best. At E they point the same way, so the curve and the line touch: E is best.

5.4 At E, the utility arrow is the price arrow stretched

At E both arrows stand square to the same line, so they point the same way: $(30, 10) = 10 \cdot (3, 1)$, that is $\nabla u = \lambda\, p$ with $\lambda = 10$.

Insight: three pictures, one condition

$$\frac{MU_x}{MU_y} = \frac{p_x}{p_y} \qquad \Longleftrightarrow \qquad \frac{MU_x}{p_x} = \frac{MU_y}{p_y} \qquad \Longleftrightarrow \qquad \nabla u = \lambda\, p.$$

Touching curves (the same slope), equal utility per euro, parallel arrows: one condition, three ways to read it.

6 The Lagrangian: the same picture in algebra

A machine that writes the arrow condition for you. The Lagrangian is utility plus λ times the money left over,

$$\mathcal{L} = u + \lambda\,(B - p_x x - p_y y).$$

Its three slopes, set to zero, are exactly the conditions we already know: slope in x: $MU_x = \lambda p_x$; slope in y: $MU_y = \lambda p_y$; slope in λ: $p_x x + p_y y = B$, the budget. No new idea: the arrow condition $\nabla u = \lambda p$, plus the budget. For our example:

$$\mathcal{L} = x\,y + \lambda\,(60 - 3x - y).$$

Set each slope to zero, one at a time:

$$\frac{\partial \mathcal{L}}{\partial x} = y - 3\lambda = 0 \qquad (MU_x = \lambda p_x)$$

$$\frac{\partial \mathcal{L}}{\partial y} = x - \lambda = 0 \qquad (MU_y = \lambda p_y)$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} = 60 - 3x - y = 0 \qquad \text{(the budget line)}$$

Solve, one line at a time:

$$x = \lambda$$

$$y = 3\lambda \qquad \text{(so } y = 3x\text{: MRS} = 3 = \text{the price ratio)}$$

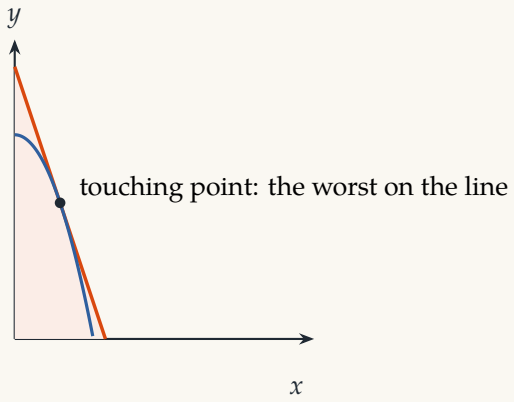
$$3\lambda + 3\lambda = 60 \quad \Longrightarrow \quad \lambda = 10$$

$$(x, y) = (10, 30)$$

The same E, and the same stretch factor as the arrows.

Insight: why the touching point is a best point

It works because the curves bow in. On a curve that bowed out, the touching point would be the *worst* bundle on the line: the two ends of the line would lie on higher curves.



7 Corners and kinks

The rule $\nabla u = \lambda p$ (or $MRS = p_x/p_y$) is for a best bundle on the line away from both ends, where the curve is smooth. At a corner it does not hold:

Back to the energy drink, with arrows. With $u = x + y$, $p_x = 3$, $p_y = 1$: $\nabla u = (1, 1)$ and $p = (3, 1)$, and each arrow stays the same at every bundle. They are never parallel, so there is no point where the free move gains 0: the free move $(1, -3)$ gains $1 - 3 = -2$, so walk toward y (utility per euro: $1/3$ on x, 1 on y). Walk to the end of the line: all on y, the corner $(0, 60)$, where $\nabla u = \lambda p$ cannot hold.

A kink. With $u = \min\{x, y\}$ (hot dogs) on $3x + y = 60$ there is no single slope at the kink, so the arrow rule cannot be used. The check is "no waste": buy on the line $x = y$, so $3x + x = 60$ and the best bundle is $(15, 15)$.

How to solve

1. Draw the budget line.
2. Draw the indifference curves.
3. Check: a corner or a kink?
4. If not: the Lagrangian, $\nabla u = \lambda p$ plus the budget.

You'll practise this in the problem set.

Reading. Frank, *Microeconomics and Behavior*, pp. 71–76 and Appendix 3A, pp. 87–92 (the utility mountain: Fig. 3A.3). Online: M. J. Osborne, *Mathematical Methods for Economic Theory* (free tutorial), §2.3 (the gradient stands at a right angle to the level curve) and §6.1 (one constraint; what λ means), `https://mjo.osborne.economics.utoronto.ca/index.php/tutorial`.